WORKED SOLUTION

STEP II 2018, Question 7

Pure • Vectors in Geometry

STEP II 2018, Question 7

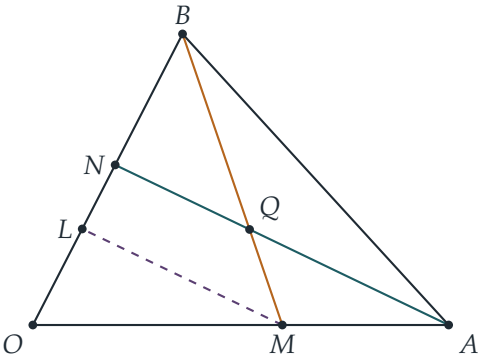
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q7

The key idea

Since M lies on OA and N lies on OB, we can write $\mathbf{m}$ as a multiple of $\mathbf{a}$ and $\mathbf{n}$ as a multiple of $\mathbf{b}$. The point Q is then reached in two ways, once along BM and once along AN, and each route gives $\mathbf{q}$ as a combination of $\mathbf{a}$ and $\mathbf{b}$. Because $\mathbf{a}$ and $\mathbf{b}$ are not parallel, we may compare coefficients, and that gives everything.

Solution



Finding $\mathbf{m}$

Write $\mathbf{m} = m\,\mathbf{a}$ and $\mathbf{n} = n\,\mathbf{b}$, where $0 < m < 1$ and $0 < n < 1$ because M and N lie on the sides.

Along BM. Q lies between M and B with $MQ : QB = \mu : 1$, so Q is the fraction $\frac{\mu}{1+\mu}$ of the way from M to B:

$$\mathbf{q} = \mathbf{m} + \frac{\mu}{1+\mu}(\mathbf{b} - \mathbf{m}) = \frac{1}{1+\mu}\,\mathbf{m} + \frac{\mu}{1+\mu}\,\mathbf{b} = \frac{m}{1+\mu}\,\mathbf{a} + \frac{\mu}{1+\mu}\,\mathbf{b}.$$

Along AN. Similarly $NQ : QA = \nu : 1$, so

$$\mathbf{q} = \mathbf{n} + \frac{\nu}{1+\nu}(\mathbf{a} - \mathbf{n}) = \frac{\nu}{1+\nu}\,\mathbf{a} + \frac{1}{1+\nu}\,\mathbf{n} = \frac{\nu}{1+\nu}\,\mathbf{a} + \frac{n}{1+\nu}\,\mathbf{b}.$$

Ask yourself Why are we allowed to equate the coefficients of **a**, and separately of **b**?

O, A and B are the vertices of a genuine triangle, so **a** and **b** are non-zero and not parallel. If $\alpha\mathbf{a} + \beta\mathbf{b} = \mathbf{0}$ with $\beta \neq 0$, then **b** would be a multiple of **a**; so $\beta = 0$, and then $\alpha = 0$. Hence each vector in the plane has only one expression as a combination of **a** and **b**.

Subtracting the two expressions for **q** and comparing coefficients:

$$\frac{m}{1+\mu} = \frac{\nu}{1+\nu}, \qquad \frac{\mu}{1+\mu} = \frac{n}{1+\nu}.$$

The first gives $m = \dfrac{(1+\mu)\nu}{1+\nu}$, that is,

$$\mathbf{m} = \frac{(1+\mu)\nu}{1+\nu}\,\mathbf{a},$$

as required. The second gives, by the same argument with the roles swapped,

$$\mathbf{n} = \frac{\mu(1+\nu)}{1+\mu}\,\mathbf{b}.$$

Finding λ

Now $\mathbf{l} = \lambda\mathbf{b}$, so

$$\overrightarrow{ML} = \lambda\mathbf{b} - m\,\mathbf{a}, \qquad \overrightarrow{AN} = n\,\mathbf{b} - \mathbf{a}.$$

These are parallel exactly when $\overrightarrow{ML} = k\,\overrightarrow{AN}$ for some scalar k. Comparing coefficients of **a** (allowed, as before) gives $k = m$, and then the **b** coefficients give $\lambda = kn = mn$. (Geometrically: $ML \parallel AN$ makes triangles OML and OAN similar, so $OL/ON = OM/OA$.) Hence

$$\lambda = mn = \frac{(1+\mu)\nu}{1+\nu} \cdot \frac{\mu(1+\nu)}{1+\mu} = \mu\nu.$$

The condition $\mu\nu < 1$

$$m < 1 \iff (1+\mu)\nu < 1+\nu \iff \mu\nu < 1,$$
$$n < 1 \iff \mu(1+\nu) < 1+\mu \iff \mu\nu < 1,$$

using $1+\nu > 0$ and $1+\mu > 0$ to clear denominators. So, given positive μ and ν, the condition $\mu\nu < 1$ is exactly what is needed for M to lie strictly between O and A and N strictly between O and B. Equivalently, the lines AN and BM (with Q dividing them in

these ratios) cross inside the triangle. It also says $\lambda < 1$: the line through M parallel to AN meets OB between O and B, which it must do, since M is on the side OA. If $\mu\nu = 1$, then $M = A$ and $N = B$, and the configuration collapses; if $\mu\nu > 1$, then M and N lie beyond A and B on the extended sides.

Checks

- *Medians.* If M and N are midpoints, then Q is the centroid, which divides each median in the ratio $2 : 1$ from the vertex. So $\mu = \nu = \frac{1}{2}$, and the formula gives $m = \frac{(3/2)(1/2)}{3/2} = \frac{1}{2}$, as it should. Then $\lambda = \frac{1}{4}$: by the midpoint theorem in triangle OAN, L is the midpoint of ON, a quarter of the way along OB.
- *Symmetry.* Swapping $A \leftrightarrow B$ swaps $M \leftrightarrow N$ and $\mu \leftrightarrow \nu$, and the formulas for m and n swap correctly. The answer $\lambda = \mu\nu$ is symmetric too.
- A computer check on about 10^5 random triangles, with random M and N on the sides, measured μ and ν and confirmed m, n and $\lambda = \mu\nu$ to within 10^{-12}. For example, $m = 0.6$ and $n = 0.55$ (as in the diagram) give $\lambda = 0.33$.

Common slips

- Getting the section ratio the wrong way round: Q is $\frac{\mu}{1+\mu}$ of the way from M to B, not from B to M.
- Equating coefficients without saying why. It relies on **a** and **b** not being parallel.
- Expecting the triangle being acute-angled to play a role. It doesn't: the argument works for any triangle.

Beyond STEP

Comparing coefficients works because **a** and **b** are *linearly independent*, so they form a basis of the plane. That is the central idea of a first linear algebra course. The neat answer $\lambda = \mu\nu$ is a ratio result in the same family as Ceva's and Menelaus' theorems, which are cleanest in barycentric coordinates. Strang's *Introduction to Linear Algebra* develops bases and coordinates carefully.

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